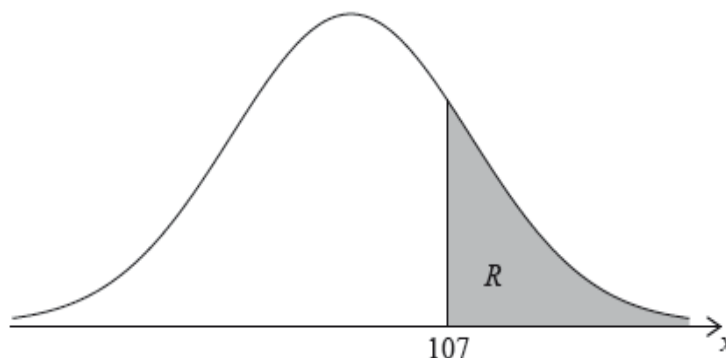


4.13 Classwork: Normal distributions

1a. The random variable X is normally distributed with a mean of 100. The following diagram shows the normal curve for X .



Let R be the shaded region under the curve, to the right of 107. The area of R is 0.24.

Write down $P(X > 107)$.

[1 mark]

1b. Find $P(100 < X < 107)$.

[3 marks]

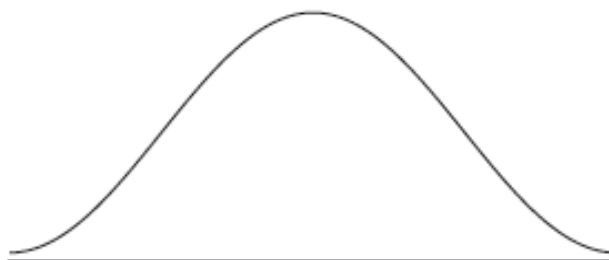
1c. Find $P(93 < X < 107)$.

[2 marks]

2a. A random variable X is distributed normally with a mean of 20 and standard deviation of 4.

On the following diagram, shade the region representing $P(X \leq 25)$.

[2 marks]



2b. Write down $P(X \leq 25)$, correct to two decimal places.

[2 marks]

2c. Let $P(X \leq c) = 0.7$. Write down the value of c .

[2 marks]

3a. The time taken for a student to complete a task is normally distributed with a mean of **20** minutes and a standard deviation of **1.25** minutes.

A student is selected at random. Find the probability that the student completes the task in less than **21.8** minutes. [2 marks]

3b. The probability that a student takes between **k** and **21.8** minutes is **0.3**. Find the value of **k** . [5 marks]

4a. The weights, **W** , of newborn babies in Australia are normally distributed with a mean **3.41** kg and standard deviation **0.57** kg. A newborn baby has a low birth weight if it weighs less than **w** kg.

Given that **5.3%** of newborn babies have a low birth weight, find **w** . [3 marks]

4b. A newborn baby has a low birth weight.

Find the probability that the baby weighs at least **2.15** kg. [3 marks]

5a. The masses of watermelons grown on a farm are normally distributed with a mean of **10** kg.

The watermelons are classified as small, medium or large.

A watermelon is small if its mass is less than **4** kg. Five percent of the watermelons are classified as small.

Find the standard deviation of the masses of the watermelons. [4 marks]

5b. The following table shows the percentages of small, medium and large watermelons grown on the farm.

small	medium	large
5 %	57 %	38 %

A watermelon is large if its mass is greater than **w** kg.

Find the value of **w** . [2 marks]

5c. All the medium and large watermelons are delivered to a grocer. [3 marks]

The grocer selects a watermelon at random from **this** delivery. Find the probability that it is medium.